Task 2 -Working with Snowflake

Create the database :

```
CREATE DATABASE IF NOT EXISTS RETAIL_DB;
```

Create the RAW schema :

```
CREATE SCHEMA IF NOT EXISTS RETAIL_DB.RAW;
```

Create the warehouse

```
CREATE WAREHOUSE IF NOT EXISTS RETAIL_WH
   WAREHOUSE_SIZE = 'XSMALL'
   AUTO_SUSPEND = 60
   AUTO_RESUME = TRUE
   INITIALLY_SUSPENDED = TRUE;

USE WAREHOUSE RETAIL_WH;

SHOW WAREHOUSES LIKE 'RETAIL_WH';
```

Create the SALES table

```
CREATE TABLE IF NOT EXISTS SALES (
   ORDER_ID NUMBER(10, 0),
   PRODUCT VARCHAR(100),
   CATEGORY VARCHAR(100),
   QUANTITY NUMBER(10, 0),
   UNIT_PRICE NUMBER(12, 2),
   CITY VARCHAR(100)
);
```

Create the CSV file format

```
CREATE FILE FORMAT IF NOT EXISTS RETAIL_CSV_FORMAT
   TYPE = CSV
   SKIP_HEADER = 1
   FIELD_OPTIONALLY_ENCLOSED_BY = '"'
   EMPTY_FIELD_AS_NULL = TRUE;
```

Create the internal stage

```
CREATE STAGE IF NOT EXISTS RETAIL_STAGE
   FILE_FORMAT = 'RETAIL_CSV_FORMAT';
```

Upload the CSV to the stage:

Upload the dataset in csv format inside the current database and RAW schema.

Load the CSV into SALES:

```
COPY INTO RETAIL_DB.RAW.SALES
FROM @RETAIL_DB.RAW.RETAIL_STAGE
FILE_FORMAT = 'RETAIL_CSV_FORMAT'
ON_ERROR = 'ABORT_STATEMENT';
```

Verify the number of records

```
SELECT COUNT(*) AS TOTAL_RECORDS
FROM RETAIL_DB.RAW.SALES;
```

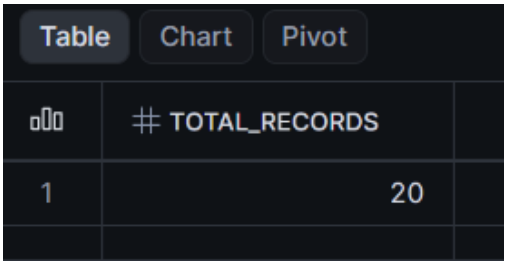
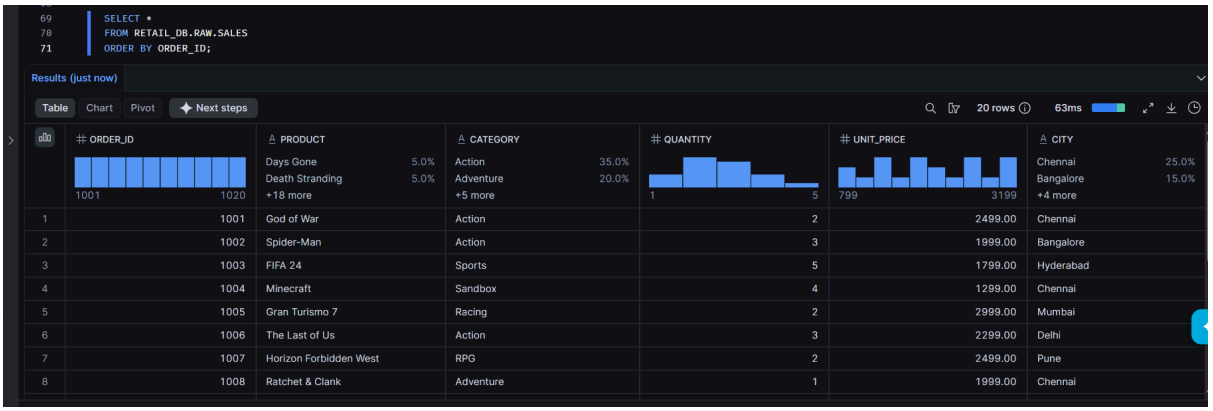


Table | Chart | Pivot

	TOTAL_RECORDS
1	20

Display all loaded records

```
SELECT *
FROM RETAIL_DB.RAW.SALES
ORDER BY ORDER_ID;
```



```
SELECT *
FROM RETAIL_DB.RAW.SALES
ORDER BY ORDER_ID;
```

Results (just now) — Table | Chart | Pivot | Next steps — 20 rows — 63ms

	ORDER_ID	PRODUCT	CATEGORY	QUANTITY	UNIT_PRICE	CITY
	1001 – 1020	Days Gone 5.0%, Death Stranding 5.0%, +18 more	Action 35.0%, Adventure 20.0%, +5 more	1 – 5	799 – 3199	Chennai 25.0%, Bangalore 15.0%, +4 more
1	1001	God of War	Action	2	2499.00	Chennai
2	1002	Spider-Man	Action	3	1999.00	Bangalore
3	1003	FIFA 24	Sports	5	1799.00	Hyderabad
4	1004	Minecraft	Sandbox	4	1299.00	Chennai
5	1005	Gran Turismo 7	Racing	2	2999.00	Mumbai
6	1006	The Last of Us	Action	3	2299.00	Delhi
7	1007	Horizon Forbidden West	RPG	2	2499.00	Pune
8	1008	Ratchet & Clank	Adventure	1	1999.00	Chennai

Games sold in Chennai

```
SELECT *
FROM RETAIL_DB.RAW.SALES
WHERE CITY = 'Chennai';
```

```
74  SELECT *
75  FROM RETAIL_DB.RAW.SALES
76  WHERE CITY = 'Chennai';
```

Results (just now)

Table | Chart | Pivot — 5 rows — 305ms

	ORDER_ID	PRODUCT	CATEGORY	QUANTITY	UNIT_PRICE	CITY
1	1001	God of War	Action	2	2499.00	Chennai
2	1004	Minecraft	Sandbox	4	1299.00	Chennai
3	1008	Ratchet & Clank	Adventure	1	1999.00	Chennai
4	1014	Sackboy	Platformer	3	1499.00	Chennai
5	1020	Infamous Second Son	Action	3	1199.00	Chennai

Quantity is greater than 3

```
SELECT *
FROM RETAIL_DB.RAW.SALES
WHERE QUANTITY > 3;
```

```
78  SELECT *
79  FROM RETAIL_DB.RAW.SALES
80  WHERE QUANTITY > 3;
```

Results (just now)

Table | Chart | Pivot — 4 rows — 70ms

	ORDER_ID	PRODUCT	CATEGORY	QUANTITY	UNIT_PRICE	CITY
1	1003	FIFA 24	Sports	5	1799.00	Hyderabad
2	1004	Minecraft	Sandbox	4	1299.00	Chennai
3	1009	Ghost of Tsushima	Action	4	2799.00	Bangalore
4	1015	Days Gone	Action	4	1399.00	Bangalore

Highest-priced games

```
SELECT *
FROM RETAIL_DB.RAW.SALES
ORDER BY UNIT_PRICE DESC;
```

```
82  SELECT *
83  FROM RETAIL_DB.RAW.SALES
84  ORDER BY UNIT_PRICE DESC;
85
```

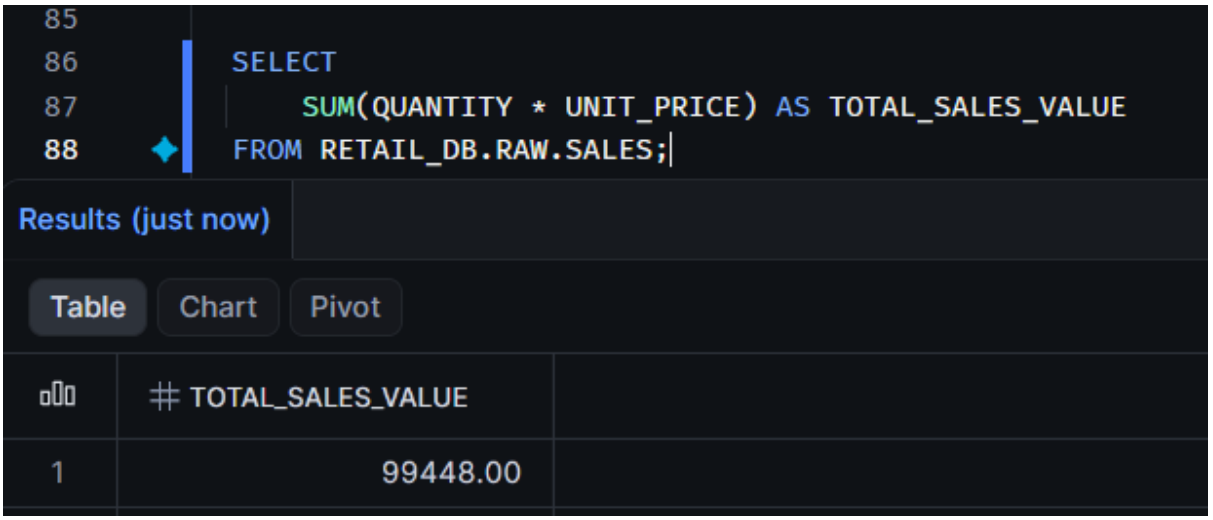
Results (just now)

Table | Chart | Pivot | Next steps — 20 rows — 64ms

	ORDER_ID (1001 – 1020)	PRODUCT (Days Gone 5.0%, Death Stranding 5.0%, +18 more)	CATEGORY (Action 35.0%, Adventure 20.0%, +5 more)	QUANTITY (1 – 5)	UNIT_PRICE (799 – 3199)	CITY (Chennai 25.0%, Bangalore 15.0%, +4 more)
1	1013	Demon's Souls	RPG	2	3199.00	Pune
2	1005	Gran Turismo 7	Racing	2	2999.00	Mumbai
3	1012	Returnal	Action	1	2999.00	Delhi
4	1009	Ghost of Tsushima	Action	4	2799.00	Bangalore
5	1001	God of War	Action	2	2499.00	Chennai
6	1007	Horizon Forbidden West	RPG	2	2499.00	Pune
7	1019	Ratchet & Clank: Rift Apart	Adventure	1	2499.00	Pune
8	1006	The Last of Us	Action	3	2299.00	Delhi
9	1002	Spider-Man	Action	3	1999.00	Bangalore

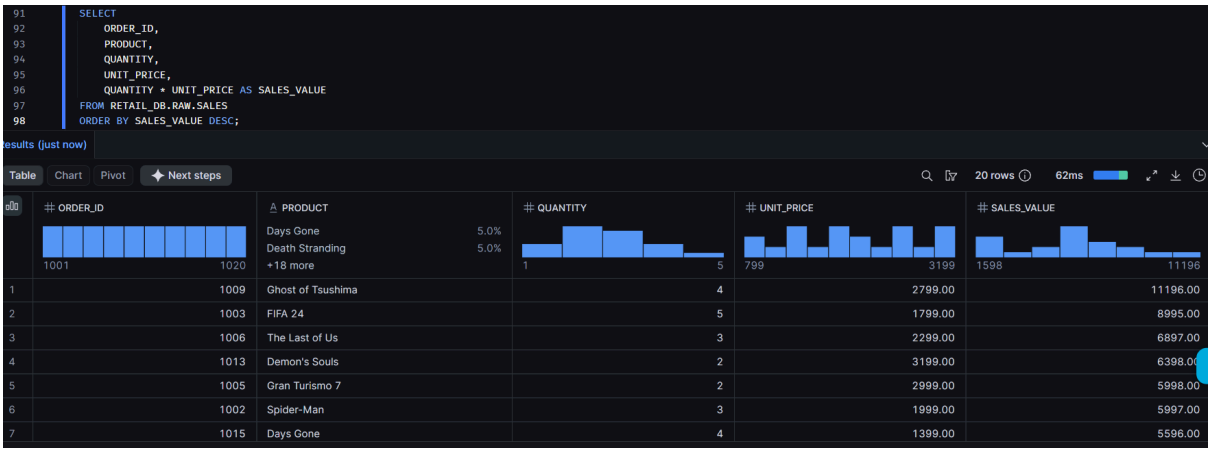
Total sales value

```
SELECT
    SUM(QUANTITY * UNIT_PRICE) AS TOTAL_SALES_VALUE
FROM RETAIL_DB.RAW.SALES;
```



Total sales value for each game:

```
SELECT
    ORDER_ID,
    PRODUCT,
    QUANTITY,
    UNIT_PRICE,
    QUANTITY * UNIT_PRICE AS SALES_VALUE
FROM RETAIL_DB.RAW.SALES
ORDER BY SALES_VALUE DESC;
```



Categories with sales value greater than ₹10,000:

```
SELECT
    CATEGORY,
    SUM(QUANTITY * UNIT_PRICE) AS TOTAL_SALES_VALUE
FROM RETAIL_DB.RAW.SALES
GROUP BY CATEGORY
HAVING SUM(QUANTITY * UNIT_PRICE) > 10000;
```

